Convolutional Neural Network for Scene Recognition

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***Abstract*—This project focuses on leveraging Convolutional Neural Networks (CNNs) and Transfer Learning to recognize various scenes from the simplified Places2 dataset, encompassing 40 distinct scene categories. By utilizing the ResNet-34 model within the torchvision framework, the project aims to achieve robust scene classification by adapting the pre-trained network to suit the specific requirements of scene recognition. Modifications include resizing images from 128×128 to 224×224 pixels, implementing data augmentations such as translations, rotations, and scalings, and refining the network architecture to enhance learning on the new scene classification task. Through systematic training, validation, and testing phases, the project evaluates the model's performance using accuracy metrics and confusion matrices, while also exploring the impact of hyperparameter tuning on the outcomes. Additional insights are gained by visually analyzing both correctly and incorrectly classified images to understand the model's strengths and limitations in real-world scenarios.**

***Index Terms*—Convolutional Neural Networks (CNN),Transfer Learning, Scene Recognition, Machine Learning, Data Augmentation, Hyperparameter Tuning**

I. Introduction

Recognizing and categorizing scenes in images is an important research topic in advanced machine learning. Scene recognition involves identifying and categorizing the environment depicted in an image, such as an airport terminal, bookstore, or playground. This capability is critical to a wide range of applications, including autonomous driving, surveillance systems, image organization, and augmented reality. Scene recognition is difficult because real-world environments are diverse and often ambiguous, with different scenes sharing similar visual features and a single scene containing multiple objects and activities. Advanced machine learning techniques, particularly Convolutional Neural Networks (CNNs) and transfer learning, have shown great promise for overcoming these challenges by improving the accuracy and robustness of scene classification systems.

In this project, we leverage CNNs and transfer learning to develop a robust scene recognition model using the ResNet-34 architecture. We utilize the Places2 simp dataset, which includes a diverse set of images across 40 scene categories. The project involves preprocessing the dataset, augmenting the training data, fine-tuning the ResNet-34 model, and optimizing various hyperparameters. The model's performance will be evaluated through standard accuracy metrics and confusion matrices, and its applicability will be tested with real-world images. This work aims to showcase the effectiveness of advanced machine learning techniques in scene recognition and provide comprehensive insights into the model's performance.

II. Literature Review

Khan et al., 2023 [1] presented a method for scene recognition by combining features from intermediate and higher layers of the Inception-ResNet-v2 architecture to capture both local and global scene structures. It achieves significant performance improvements over existing methods on various datasets, with accuracies ranging from 80.31% to 99.05%. However, further optimization is needed for indoor scene datasets due to high inter-class similarity, and its suitability for real-time applications is limited by the time-consuming ensemble of classifiers. Nonetheless, the method shows good results for object recognition tasks.

Arrun Sivasubramanian et al., 2023 [2] presented a method for identifying scenes in remote sensing photographs, which is useful for tasks such as land use and satellite analysis. It analyzes different deep learning designs, including as convolutional neural networks (CNNs) like VGG-16 and ResNet50, as well as transformer architectures like Vision Transformers (ViT) and Swin Transformers. By fine-tuning these models with pre-trained weights and a LinBnDrop layer, they enhance accuracy and F1-score on benchmark datasets. The trials demonstrate up to a 2% increase in accuracy, with some models reaching 100%. These findings demonstrate the efficacy of the suggested approach for scene categorization tasks, providing important insights for researchers and applications in land use and satellite analysis.

Pinhe Wang et al., 2022 [3] introduced an improved method for recognizing urban scenes using Convolutional Neural Networks (CNNs). It addresses issues like image impurities and low quality by applying a preprocessing technique based on Canny edge detection. The model's stability is enhanced using the L2 regularization algorithm, improving its ability to recognize scenes accurately. By collecting datasets from different environmental conditions, experiments confirm the

method's effectiveness, achieving an accuracy of 84.5% on the NUC dataset. Furthermore, an image detection algorithm based on saliency helps highlight important features, improving recognition accuracy to 95.3%. The method outperforms other networks like PlusNet and HybridNet, showing promise for future research to enhance recognition accuracy and speed further.

Pui Sin Yee et al., 2022 [4] introduced DeepScene, a scene classification method employing Convolutional Neural Networks (CNNs) with spatial pyramid pooling. Addressing challenges like shared components and varying similarity within classes, DeepScene integrates CNNs with spatial pooling to handle diverse object sizes and positions. Additionally, it proposes methods for converting grayscale images to RGB, crucial for scene classification. Through ensemble learning and knowledge transfer from large-scale datasets, DeepScene achieves high accuracy on Event-8 (98.1%), Scene-15 (95.6%), and MIT-67 (71.0%).

Sarfaraz Masood et al., 2020 [5] were focussing on scene recognition using Convolutional Neural Networks (CNNs) and evaluates different models on the PLACES2 dataset, containing millions of labeled scene images. The study uses AlexNet, VGG-16, and Inception v3 models to classify scenes, achieving average accuracies of 79.8%, 87.3%, and 89.3%, respectively. In particular, Inception v3 performs well, with test accuracies of 94.6% and 85.7% on SUN397 and NUS-WIDE datasets, surpassing previous results. The findings suggest that Inception v3 is effective for scene recognition tasks, especially with large image datasets.

III. Methodology

A. Dataset

The dataset used for scene recognition is called Places2_simp. It's a smaller version of the Places2 dataset, which is massive, with over 10 million images of more than 400 different types of scenes. In Places2_simp, there are 40 types of scenes, and each type has 1000 images, making it a total of 40,000 images. When it comes to splitting the dataset, 80% of the images are for training, and 20% are for testing. However, during testing, some images were missing, so we gathered more to complete the test set. Each scene category in the test set has 10 images, totaling 400 images. These images in the test set are specifically based around areas like Guildford, Surrey, and London.

B. Data Augmentation

Data augmentation is a crucial strategy employed to enhance the model's ability to generalize by artificially enlarging the training dataset through various transformations. The transformations applied in this project include:

- Resizing: All images are resized to 224x224 pixels to ensure uniformity in image size, which is essential for processing through the neural network.
- Grayscale Conversion: Images are converted to grayscale (single-channel), reducing the complexity of the model by focusing on texture and shape instead of color.
- Random Horizontal Flip: During training, images are randomly flipped horizontally. This transformation mimics the scenario where objects appear in reversed orientation, increasing the robustness of the model.
- Random Rotation: Images are randomly rotated within a range of 20 degrees. This introduces variability in the orientation of objects, aiding the model to recognize objects across different angles.
- Random Affine Transformation: Affine transformations with no rotation, translations up to 5% of the image dimensions, and scaling between 90% and 110% are applied. These transformations simulate changes in object scale and position within the scene, further enhancing the diversity of the training data.
- Normalization: Pixel values are standardized across all datasets (training, validation, test) based on a mean of 0.485 and a standard deviation of 0.229, assuming the images are grayscale. Normalization helps in speeding up the convergence during training by ensuring that input features are on a similar scale.
- Tensor Conversion: All images are converted into PyTorch tensors to facilitate operations within the PyTorch framework.

C. Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are a specialized type of deep neural networks extensively used for processing structured array data like images. Their distinct architecture is specifically designed to effectively manage pixel-based data. Typically, a CNN includes several layers: convolutional layers that perform the primary data processing, pooling layers that reduce data dimensionality, and fully connected layers that compile and interpret the features extracted by prior layers.

The activation f_{ij}^l at position (i, j) in the l-th layer is given by:

$$f_{ij}^l = \sigma\left(\sum_m \sum_n \sum_k w_{mnk}^l \cdot x_{i+m,j+n,k}^{l-1} + b^l\right) \quad (1)$$

where:

- σ denotes an activation function such as ReLU,
- w_{mnk}^l represents the weights of the filter at layer,
- $lx_{i+m,j+n,k}^{l-1}$ is the input from the previous layer, and
- b^l is the bias.

1) ResNet: Residual Networks, commonly known as ResNet, represent a pivotal advancement in neural network technology designed to overcome the issue of vanishing gradients through the use of skip connections. These connections act as shortcuts that bypass one or more layers. Traditional CNNs often struggle with increased depth and complexity, which can hinder effective training due to vanishing gradients. ResNets solve this challenge with a straightforward but effective solution: introducing shortcut connections that allow some layers to be skipped, facilitating more efficient training even in deeper network architectures. ResNet34 is a variant of ResNet which consists of 34 layers. The architecture is composed

of several residual blocks, each containing a sequence of layers and a shortcut connection that bypasses Initial Layers, Residual Blocks and Ending Layers.

$$y = F(x, \{W_i\}) + x \tag{2}$$

where:

- x and y are the input and output vectors of the layers considered,
- $F(x, \{W_i\})$ represents the residual mapping to be learned, and
- For the layer outputs to match the dimensions of x, F plus the input x must be the same dimensions, which is achieved through either dimension matching or projection shortcuts.

D. Model Configuration and Architecture

a) Model Configuration:

- Batch Size: 12,
- Number of Epochs: 8,
- Optimizer: AdamW,
- Learning Rate: 0.0001,
- Weight Decay: 0.0005,
- Scheduler: CosineAnnealingLR,

b) Model Architecture:

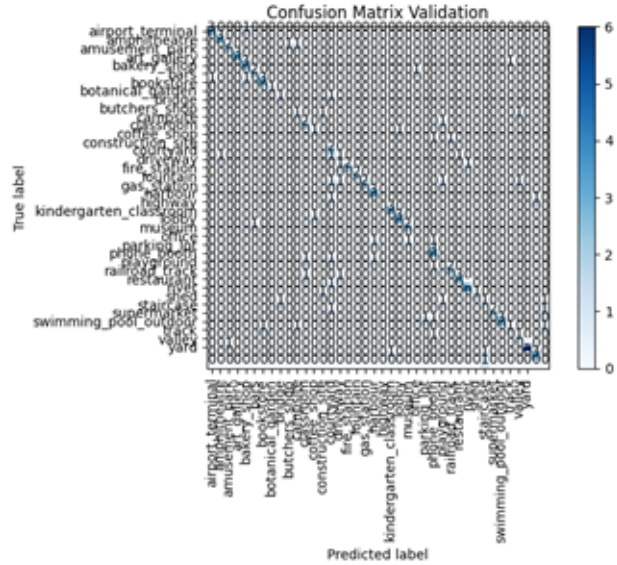
- Convolution Layer 1: Customized for single-channel input (nn.Conv2d(1, 64, kernelsize=7, stride=2, padding=3, bias=False)),
- FC1: nn.Linear (from ResNet's infeatures to 256),
- Activation: nn.ReLU,
- Dropout: nn.Dropout(0.4),
- FC2: nn.Linear (256 to 40),

RESULTS AND ANALYSIS

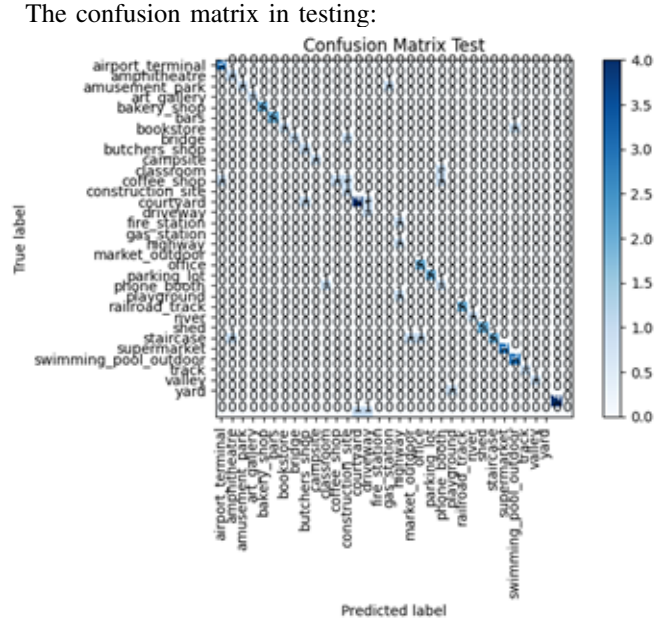
The Pre-trained ResNet34 model was trained to classify the data into 40 categories, each category containing 100 images. In total the training images were 40000 images whereas the testing dataset was with a merger 400 images, with each class having 10 images. On training the model for 8 epochs we found the following observations:

We see that at training loss is reducing continuously till the last epoch which shows the training is not over-fitting but we see that validation loss is least in the 6th epoch and increases slightly in the next epochs which suggests that the model is begins to over-fit. The same can be seen in the validation loss and training loss graphs below. The final validation accuracy of the trained model is 51.225% in the last epoch and the top-5 and top-1 accuracies in last epoch are 81.62% and 51.24% respectively. In testing we get a better overall testing accuracy and better top-5 accuracy with 66.908% and 93.72% respectively.

Below graphs for validation accuracy, top-k accuracies suggest that training the model further will lead to over-fitting. From the below confusion matrix and classification report for testing we see that the model has achieved a good accuracy and good f1 score for most of the categories. The good recall value and precision also suggests that the model is able to distinguish the categories and identifying them. The following are the confusion matrices obtained by training the Pre-trained Resnet34 model.



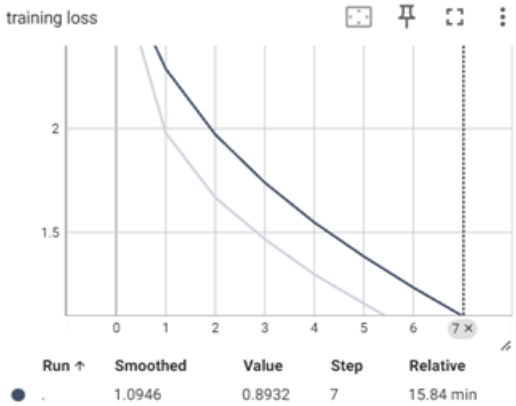
The confusion matrix in testing:

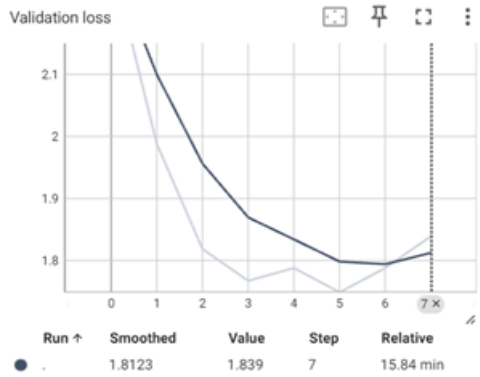


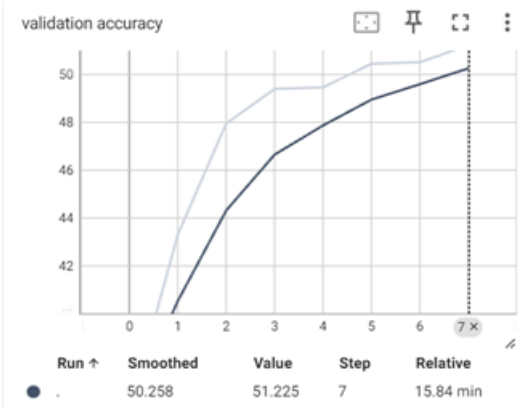
The classification report for testing the model:

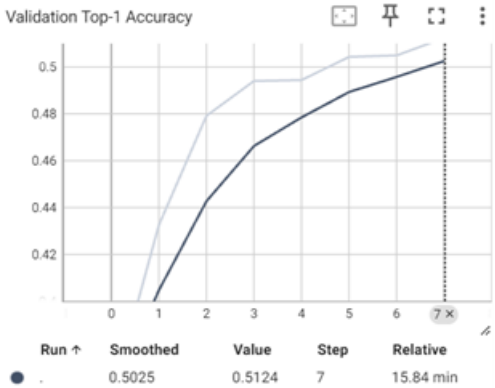
	precision	recall	f1-score	support
amphitheatre	1.00	1.00	1.00	2
amusement_park	0.50	1.00	0.67	1
art_gallery	1.00	1.00	1.00	2
bakery_shop	1.00	1.00	1.00	1
bars	0.00	0.00	0.00	0
bookstore	1.00	0.33	0.50	3
botanical_garden	0.00	0.00	0.00	0
bridge	0.00	0.00	0.00	1
bus_interior	1.00	1.00	1.00	2
butchers_shop	0.00	0.00	0.00	1
campsite	1.00	1.00	1.00	1
classroom	0.57	1.00	0.73	4
coffee_shop	0.00	0.00	0.00	2
construction_site	0.50	0.50	0.50	2
courtyard	0.00	0.00	0.00	0
driveway	0.75	1.00	0.86	3
fountain	1.00	0.67	0.80	3
harbour	0.67	1.00	0.80	2
highway	1.00	0.67	0.80	3
kindergarten_classroom	1.00	0.25	0.40	4
lobby	1.00	1.00	1.00	3
market_outdoor	1.00	1.00	1.00	1
museum	0.00	0.00	0.00	1
office	0.50	1.00	0.67	1
parking_lot	0.67	1.00	0.80	2
phone_booth	1.00	1.00	1.00	2
playground	1.00	1.00	1.00	2
railroad_track	0.00	0.00	0.00	0
restaurant	0.00	0.00	0.00	0
river	0.50	0.50	0.50	2
staircase	1.00	1.00	1.00	3
supermarket	0.00	0.00	0.00	0
swimming_pool_outdoor	1.00	1.00	1.00	3
track	1.00	0.50	0.67	2
valley	1.00	0.75	0.86	4
yard	0.00	0.00	0.00	1
accuracy			0.73	64
macro avg	0.60	0.59	0.57	64
weighted avg	0.80	0.73	0.73	64

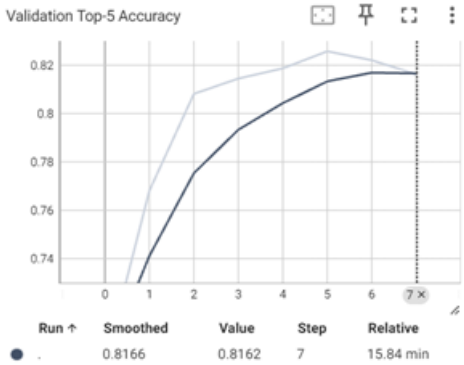
The tensor-board plots for train loss, validation loss, validation accuracies including top-k:











Conclusion

In this experiment, we attempted at using deep neural networks with Convolutional Neural Networks as backbone with custom Fully Connected layer for scene recognition. This network combination resulted in an overall accuracies for validation and test are 51.22% and 66.908%. This shows that the model has preformed well and one of the suitable models for the task of scene recognition based on images. This could be implemented in real life application like to help blind people navigate through places using smart devices, surveillance cameras, drones,etc.. These models can further be enhanced by training with larger datasets with more classes and images.

Future Works

Several promising directions exist for future work to enhance the effectiveness and applicability of scene recognition systems. Incorporating more advanced data augmentation

techniques, such as adversarial training, can create more diverse and representative training samples, helping the model generalize better to unseen data. Additionally, methods like mixup and CutMix, which combine multiple images to generate new training samples, as well as grid masking, which randomly masks out parts of the image, can further enhance the model's ability to handle complex scenes.. Implementing self-supervised learning methods can reduce reliance on large labeled datasets by leveraging vast amounts of unlabeled data, leading to improved feature learning and efficiency.

Exploring attention mechanisms and transformer models can improve accuracy by allowing the model to concentrate on the most relevant parts of an image, resulting in better feature representation.Fine-tuning hyperparameters and experimenting with different CNN configurations can help the model run more efficiently and accurately. These future directions seek to improve and optimize CNN-based scene recognition systems, making them more accurate and reliable.

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